1) find out all the list of companies which are conducting placement drive using DSA

Google, Microsoft, Amazon, Flipkart, zomato, swiggy, paytm, phonepe, TCS, Infosys, Wipro, cognizant, IBM, Dell, Meesho.

2) what is the syllabus of DAS in GATE

Data structures: Arrays, stacks & queues, linked lists (single, Double, circular), Trees & Binary Trees, BST, traversals, in order/ pre order/ post order), Binary search Trees. Binary search Trees, Binary Heaps, Graphs

Algorithms:- sorting, searching, Asymptotic worst-case time & space complexity, Algorithm design Techniques, Graph algorithms.

3) find out what are the set of variables in Amazon ← Invoice Bill.

① string customername
② Int orderNumber
③ String orderdate
④ String shippingAddress
⑤ String Deliveryplace
⑥ String productDescription
⑦ float unit price
⑧ Double NetAmount
⑨ float TaxRate
⑩ String Taxtype
⑪ Double Total Amount of Each product
⑫ String Amount in words
⑬ Int No of Items

4) find out what are the set of variables in Gasbill and

Gas Bill

long consumerNO

String Name

String Address

String Int orderNo

String orderDate

float prise

float CGST

float SGST

Double finalprice

String orderstatus

Int Issue date

current Bill

long BillNo

Int units

long Amount

Int last paidDate

long Amountdue

Int Time

String SECName

long USCNO

String Name

String Address

1. Applications of Recursion

- Tree traversal (Binary trees, BSTs)
- Graph traversal (Depth-first search - DFS)
- Divide and conquer Algorithms (Merge sort, Quick sort, Binary search)
- Dynamic programming (Memoization)
- Backtracking problems (N-Queens, Sudoku solver, Maze solving)
- Dynamic programming (Memoization)
- file and folder traversal in operating systems.
- parsing expressions in compilers.
- Mathematical computations (factorial, fibonacci, GCD)

2. Is Recursion still used in the present programming era? If yes, where?

Yes. Recursion is still widely used in modern programming because it provides simple and elegant solutions to many problems.

It is commonly used in:

- Web Browsers (processing HTML/XML document trees)
- compilers and interpretors
- Artificial Intelligence (AI) and Machine learning algorithms.

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- Database query processing
- file system navigation
- Game development (pathfinding and decision trees)
- Data structures like trees and graphs.

Although Recursion is useful; developers sometimes replace it with loops or stacks for better performance and to avoid stack overflow in very deep recursion.

3. What are the languages that support Recursive calls?

- C, C++, Java, Python, JavaScript, C#, Kotlin, swift, Go, Rust, PHP, Ruby, Scala, Haskell, Lisp.

⇒ find out the History of Tower of Hanoi

The Tower of Hanoi is famous mathematical puzzle invented by Edouard Lucas in 1883

Lucas introduced the puzzle under the name "The Tower of Brahma" and described a legend about monks in an Indian temple.

According to the legend:

- There are three rods and 64 golden disks stacked on one rod, with the largest at the bottom and the smallest at the top
- The monks move one disk at a time while following these rules:
  a. only one disk may be moved at a time
  b. A large disk can never be placed on top of a smaller disk
  c. only the top disk of any rod can be moved.
- The legend says that when all 64 disks are moved to another rod, the world will come to an end. This is a fictional story created to make puzzle more interesting.

Today, the Tower of Hanoi is widely used in computer science and mathematics to teach recursion, algorithm design, and problem-solving. It is one of the most popular examples for understanding recursive.

⇒ List out of the Data structures you addressed in your surroundings

Tooth Brush Holder	Array
plate stack	Stack
Breakfast queue	Queue
phone contacts	Hash Map
classroom seating	2D-Array
college attendence list	Linked list
college timetable	Array
file folders on laptop	Tree
college campus roads	Graph
Browser Back Button	stack
music playlist	Linked List
Bus queue	Queue

H.W Java Generates Hashcode for every object find out what is the Hashcode in Java. And the purpose of Generating the Hash code.

- Hash code is an integer value returned by the hashcode() method
- every Java object has a hash code
- By default, java generates it using the object class (based on object identity)
- Some classes (like String) generate hash codes from their contents
- purpose: To improve the speed of searching, storing and retrieving objects in collections such as HashMap, HashSet and Hashtable
- equal objects must have the same hash code.

HW 1: What are the majority problems that you can observe in our surroundings
Pick any 5 problems. find out they can be solved by

DSA

1. Traffic Navigation

problem: finding the shortest route to a destination.

DSA used: Graphs, Dijkstra's Algorithm.

Solution: find the shortest and fastest path, reducing travel time and fuel usage

2- Hospital patient Management

problem: Managing patients based on emergency Level

DSA used : priority queue (Heap)
Solution: Treats Gitical patients first or faster Emergency care.

3. Library Book search
problem: Searching for a Book among thousands of Books
DSA used: Binary search, Hash Table
Solution: finds Books quickly using Book ID or title

4. Social Media friend Suggestions
problem: Recommending friends with common interests.
DSA used: Graph.
Solution: connectors user through multiple friends and shared interests

5. online shopping product search.
problem: quickly finding products from a large catalog.
DSA used: Hash Table, Trie, sorting and searching algorithms
Solution: displays relevant products instantly based on user search.